INDIA.RUNS

Build what next India runs on

India.Runs - Redrob Candidate Ranking

- Team Name : india-runs-data-ai
- Team Leader Name : Mohamed Fasidh
- Problem Statement : Build an offline AI system that ranks candidates for Redrob's Senior AI Engineer founding-team role by understanding production ML fit, behavioral availability, logistics, and profile consistency beyond keyword matching.

INDIA.RUNS

Build what next India runs on

Solution Overview

- Proposed solution: an offline deterministic ranker that streams the 100K candidate JSONL, extracts JD-aligned signals, scores every candidate, and writes a validated top-100 CSV.
- It is designed for the exact Redrob JD: Senior AI Engineer with production retrieval, ranking, vector search, evaluation, Python, product-engineering, and India/Pune/Noida fit.
- Differentiator: the ranker does not count AI keywords. It combines career evidence, skill trust, Redrob behavior, logistics, education, and honeypot/profile consistency checks.
- Every selected candidate receives a grounded 1-2 sentence reason using only facts present in the candidate profile.

JD Understanding & Candidate Evaluation

- JD requirements extracted: 5-9 years preferred, production embeddings/retrieval, vector or hybrid search, ranking evaluation (NDCG/MRR/MAP/A-B tests), strong Python, and product-company shipping judgment.
- Positive career signals: applied ML/AI/search/recommendation titles, shipped production ranking/search systems, product or marketplace company exposure, recent hands-on coding, and mentoring ability.
- Negative fit signals: pure research without deployment, recent-only LangChain demos, services-only career history, non-technical current titles with AI keyword stuffing, and weak NLP/IR exposure.
- Behavioral signals: open_to_work, recent activity, recruiter response rate, notice period, interview completion, GitHub activity, verification, recruiter saves, and relocation/hybrid readiness.

Ranking Methodology

- Retrieval/ranking: all candidates are scored in one streaming pass, then sorted by score and candidate_id for deterministic output.
- Skill component: Python, retrieval/RAG, vector databases, ranking/recommendations, evaluation, LLM/fine-tuning, and production system evidence with endorsement/duration/assessment trust.
- Career component: current title, experience band, role descriptions, product-company exposure, services-only penalties, shipped search/ranking/recommendation evidence, and research/demo-only penalties.
- Final score: weighted combination of skill (30%), career (34%), behavior (16%), logistics (12%), education (8%), minus honeypot penalties.

Explainability & Data Validation

- Explanations are generated from candidate fields already loaded by the scorer: title, years, location, matched skill groups, top profile skills, response rate, notice period, and open_to_work.
- No hosted LLM is used for reasoning, so justifications cannot invent employers, skills, or achievements outside the profile.
- The tone changes by rank bucket: strong fit, good fit, or borderline fit. Obvious concerns such as long notice, inactivity, weak relocation, or consistency flags are included.
- Suspicious profiles are downweighted for expert skills with near-zero duration, broad skill lists with low experience, AI keyword-heavy non-technical titles, and timeline inconsistency.

End-to-End Workflow

- 1. Read the Senior AI Engineer JD and convert it into explicit feature groups.
- 2. Stream each candidate JSON object from candidates.jsonl or candidates.jsonl.gz.
- 3. Build a profile text blob from profile, career history, skills, education, certifications, and Redrob signals.
- 4. Score skills, career, behavior, logistics, education, and honeypot risk.
- 5. Sort candidates by final score and deterministic candidate_id tie-break.
- 6. Write candidate_id, rank, score, and grounded reasoning to submission.csv.
- 7. Validate using the official validate_submission.py script.

System Architecture

- Input Layer: candidates.jsonl + Redrob JD + candidate schema.
- Feature Layer: text evidence, skill trust, career history, Redrob behavior, logistics, education, and consistency checks.
- Scoring Layer: weighted interpretable components and honeypot penalties.
- Ranking Layer: deterministic sort, strictly decreasing displayed scores, top-100 selection.
- Output Layer: validated submission.csv, metadata YAML, README, GitHub repo, and Colab sandbox demo.

Results & Performance

- Generated the required 100-row submission.csv and passed the official validator: 'Submission is valid.'
- Full 100K-candidate run completed locally in about 54-100 seconds, under the 5 minute CPU-only challenge limit.
- Memory footprint is small: the script streams input and stores compact score rows before sorting.
- Ranking quality is driven by JD-specific evidence rather than generic similarity: top candidates show retrieval/ranking/vector/evaluation production signals plus availability.
- Colab sandbox demo runs successfully on demo_candidates.jsonl and prints 'Demo ranking generated successfully.'

Technologies Used

- Python standard library for ranking: json, csv, gzip, argparse, datetime, pathlib, math, and regex.
- No external model, GPU, or network dependency during ranking, matching the reproduction constraints.
- Pandas is used only in the Colab demo to display the generated demo_submission.csv.
- Matplotlib/lxml are used only for documentation/deck generation, not for ranking.
- GitHub hosts source code and artifacts; Google Colab provides the sandbox/demo notebook.

INDIA.RUNS

Build what next India runs on

Submission Assets

- GitHub repository: https://github.com/Mohamed-Fasidh/redrob-candidate-ranker
- Sandbox/demo: https://colab.research.google.com/github/Mohamed-Fasidh/redrob-candidate-ranker/blob/main/redrob_ranker_demo_colab.ipynb
- Submission CSV: submission.csv (validated with official script)
- Reproduce command: python rank.py --candidates ./candidates.jsonl --out ./submission.csv
- Contact: Mohamed Fasidh | mohamedfasidh045@gmail.com | +91-8015334576

Thank You

- Thank you
- The solution package is complete, reproducible, offline, and ready for Redrob review.
- Team: india-runs-data-ai